

YallaFix

Smart Campus Facility Management System

Setup & Installation Guide

1. Overview

YallaFix is a mobile application built for the German International University (GIU) to streamline campus facility issue reporting and resolution. The system connects three types of users: Community Members (reporters), Facility Managers, and Workers.

2. Technology Stack

Layer	Technology
Frontend	React Native (Expo SDK 54)
Backend	Node.js with Express.js
Database	PostgreSQL (Supabase)
Authentication	JWT (JSON Web Tokens) + Bcrypt
Storage	Supabase Storage (Cloud Object Storage)
Version Control	Git & GitHub

3. Prerequisites

- Node.js v18 or higher
- npm v9 or higher
- Expo CLI: `npm install -g expo-cli`
- Expo Go app (download from iOS App Store or Google Play)
- Supabase account (free tier available)
- Git installed on your machine

4. Project Structure

YallaFix/

- app/ - Expo Router screens
- └─ (tabs)/ - Tab navigator
- └─ auth/ - Login, Register, Forgot Password
- └─ community/ - Community member screens
- └─ manager/ - Facility manager screens
- └─ worker/ - Worker screens
- └─ _layout.tsx - Root layout + auth routing backend/ - Node.js + Express server
- └─ src/
- └─ routes/ - API route handlers

- | | — middleware/ - Auth, role validation
- | | — db.js - Database connection
- | | — index.js - Express entry point
- | — .env - Environment variables

5. Installation & Setup

5.1 Clone the Repository

git clone https://github.com/nadeenkhalifa/YallaFix cd
YallaFix

5.2 Set Up the Backend

1. Navigate to backend folder: cd backend
2. Install dependencies: npm install
3. Create .env file with these variables:

```
DATABASE_URL=DATABASE_URL=postgresql://postgres.ruzakhretrdgdoybruws:Genan%40258200
@aws-0-eu-west-1.pooler.supabase.com:6543/postgres?pgbouncer=true
JWT_SECRET=yallafix_secret_key_2026
NODE_ENV=development
PORT=3000
```

5.3 Set Up the Frontend

6. From project root (go back to YallaFix/):
cd ..

7. Install dependencies:
npm install

8. Install Async Storage: npx expo install @react-native-async-storage/async-storage

9. Update API base URL in services/api.ts:
const BASE_URL = 'http://YOUR_PC_IP:3000/api';

- Replace YOUR_PC_IP with your machine's IP address
- Find IP: Windows (ipconfig) | Mac/Linux (ifconfig) • Ensure phone and PC are on same WiFi network

10. Start the frontend: npx expo start

11. Scan QR code with Expo Go app on your phone

- iOS: Open Camera → tap notification
- Android: Open Expo Go app → Scan QR

6. Database Setup

Run the following SQL in Supabase SQL Editor:

-- Create tables (see DATABASE_SCHEMA documentation for full schema) -- Or
execute the provided database migration scripts

7. User Roles & Permissions

Role	Permissions
Community Member	Submit issues, view own issues, confirm issues, comment
Facility Manager	View all issues, assign workers, change status, post updates
Worker	View assigned issues, update progress, upload proof

8. Running the Application

8.1 Start Backend

cd backend && node src/index.js

Expected: Server listening on http://localhost:3000

8.2 Start Frontend

cd .. && npx expo start

Scan QR with Expo Go on your phone

8.3 Test the App

- Create an account with role 'reporter'
- Submit a complaint with photo, location, category
- Login as 'manager' role to view and assign
- Login as 'worker' to view assigned tasks

9. Troubleshooting

9.1 'Cannot reach server' or '404 errors'

- Verify backend is running on port 3000
- Check API base URL in frontend/services/api.ts
- Ensure phone and PC are on same WiFi

9.2 Database connection fails

- Verify DATABASE_URL in backend/.env
- Check that @ symbol is URL-encoded as %40
- Test connection with: node -e "const pool = require('pg').Pool; ..."

9.3 Permission errors (403)

- Verify user role matches API requirements (e.g., manager for assign-worker)
- Check JWT token is valid and included in Authorization header

10. Team

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Name	ID	Role
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